

String, StringBuilder, StringBuffer - Exercises

Assignment 1: Reverse a String using all 3 approaches

- Take a string input from the user.
- Reverse it using:
 - String (using loop)
 - StringBuilder (using reverse())
 - StringBuffer (using reverse())

Assignment 2: Concatenate Strings

- Create a list of 3 strings.
- Concatenate them using:
 - + operator
 - StringBuilder.append()
 - StringBuffer.append()
- Measure and compare performance using System.nanoTime().

Assignment 3: Apply String methods

- Take input like " Java Programming " and perform the following:
 - trim()
 - toLowerCase(), toUpperCase()
 - replace(" ", "-")
 - substring(), indexOf("a")

Assignment 4: Analyze a sentence

- Given "Learning Java is fun and interesting", do:
 - Count number of words using split()
 - Find first and last occurrence of "i" using indexOf() & lastIndexOf()
 - Check if it starts with "Learn" and ends with "ing"

Assignment 5: Demonstrate immutability

Create a string:

```
String s1 = "Hello";
```

```
String s2 = s1;
```

```
s1 = s1.concat(" World");
```

- Print both s1 and s2 to show that the original string wasn't changed (immutability).

Assignment 6: Show why immutable strings are safe

- Ask user for a password.
- Create a method that "encrypts" the password.
- Show that using immutable strings ensures the original password doesn't change after encryption.

Assignment 7: Compare pooled vs non-pooled strings

```
String s1 = "Java";
```

```
String s2 = "Java";
```

```
String s3 = new String("Java");
```

- Use == and equals() to compare s1, s2, s3 and explain what happens.

Assignment 8: Interning

```
String a = new String("Hello");
```

```
String b = a.intern();
```

```
String c = "Hello";
```

- Compare a == b, b == c, a == c and explain how intern() works.

Assignment 9: Case-sensitive and case-insensitive comparison

- Input 2 strings (e.g., "Java" and "java")
- Compare using:
 - equals()
 - equalsIgnoreCase()
 - compareTo()

Assignment 10: Sorting Strings

- Input 5 names into an array.
- Sort them using Arrays.sort() and display in:
 - Ascending order
 - Descending order